

ClipQuest

Turning public YouTube lessons into active, personalized learning experiences

CURRENT STATUS

Concept and implementation plan only. As of 19 August 2026, the repository contains a README and an Apache 2.0 license, but no application code has been written. Every technical capability in this document describes what our team plans to build.

Team leader	UnoxyRich
Team name	队长cos男娘去跟网恋对象吃饭被骗五百块
Team handle	@404-Waifu-Not-Found/cos
Members	UnoxyRich, JimmyfaQwQ, ILikeLayla (Layla), Justin-Yonardo
Project	ClipQuest
Submission	HHCC 2026
Repository	github.com/404-Waifu-Not-Found/HHCC2026
Team page	github.com/orgs/404-Waifu-Not-Found/teams/cos

This document intentionally uses future-tense language. It records our shared product direction, planned architecture, expected challenges, and intended division of work without claiming that a prototype or production system already exists.

ONE-SENTENCE VALUE

ClipQuest will transform public YouTube lessons into interactive quizzes so that learners can replace passive watching with active recall, immediate feedback, and visible progress.

We chose this theme because online video is one of the easiest ways to access explanations, demonstrations, and revision material, yet watching alone does not guarantee understanding. A learner can finish a long lesson, feel familiar with the topic, and still be unable to recall the key ideas without prompts. Creating a useful quiz manually takes additional time, and many students do not know which concepts deserve attention. Existing study tools also often require copying notes into another platform, which interrupts the learning flow.

ClipQuest is planned as a bridge between accessible video content and evidence-based study habits. Our goal is to reduce the effort required to begin active recall while preserving learner control. The product will focus on public YouTube lessons, use available captions as the grounding source, and let the learner decide when a lesson should become a quiz. By keeping the learner's AI credential on the device, we also plan to avoid placing a private API key in our backend. This direction responds to a practical need: students want quick, focused practice that remains connected to what they just watched.

A learner will open a supported YouTube lesson and launch ClipQuest through a Chrome extension or, later, a native mobile app. The product will retrieve or receive the video's captions, divide the lesson into manageable concept segments, and ask an AI model to generate grounded questions. The first validated question is intended to appear as soon as it is ready so that the learner can start before the full quiz finishes generating.

- Generate multiple-choice, true/false, and short-answer questions from caption evidence.
- Show one question at a time, accept an answer, and provide clear feedback linked to the lesson content.
- Track completion and performance, save lessons, and support later review of difficult concepts.
- Share consistent quiz data contracts across the web, Chrome extension, Android, and iOS experiences.
- Keep the learner's DeepSeek API key on the device while using the backend only for authentication and stored progress.

The interaction will be designed around a short path: choose a lesson, begin the generated quiz, answer and review, then return to saved material. If captions are missing or generation fails, the interface will explain the problem and offer a recovery action instead of leaving the learner at a blank screen.

How We Plan to Build It

No implementation exists yet. We plan to organize ClipQuest as a TypeScript monorepo so that web, extension, mobile, API, and shared validation packages can evolve together. React 19 will support the web interface, while React Native with Expo will provide the planned Android and iOS applications. A Chrome extension will handle the browser-side lesson workflow and local AI-key storage. The extension or native client will send caption-grounded prompts directly to DeepSeek using the learner's own key, rather than routing that credential through our servers.

For server-side responsibilities, we plan to use Cloudflare Workers with Hono for a small API, D1 for relational storage, and Better Auth for identity and session management. Zod schemas will validate quiz imports and keep question, answer, feedback, and progress records consistent across clients. Generation will use caption chunking, prompt constraints, schema validation, duplicate detection, and answer-consistency checks. We will accept only complete, valid questions and will retry missing items rather than discarding already accepted work.

PLANNED STACK

TypeScript; React 19; React Native; Expo; Chrome Extensions; DeepSeek API; Cloudflare Workers; Hono; D1; Better Auth; Zod; Vitest; Playwright.

Testing will be planned from the beginning. Vitest will cover pure logic and shared contracts, while Playwright will exercise real browser flows such as opening a lesson, generating a first question, answering a full quiz, and recovering from errors. We will also test on Android and iOS simulators and, when available, real devices. Security checks will verify that credentials remain local, user-owned records are access-controlled, and malformed or oversized quiz imports are rejected.

Individual Contributions

The GitHub team @404-Waifu-Not-Found/cos has four members. Because the repository is still at the planning stage, the division below records intended individual ownership rather than completed implementation. Responsibilities may overlap during integration and testing.

- **UnoxyRich** - team leader and integration owner: coordinate scope and milestones, maintain the shared data contracts, review cross-platform decisions, integrate workstreams, and lead submission documentation and demonstrations.
- **JimmyfaQwQ** - extension and AI workflow owner: develop the planned Chrome extension, obtain caption data, manage local DeepSeek credentials, design grounded prompts, stream quiz results, and add validation and recovery logic.
- **ILikeLayla (Layla)** - web and product design owner: design the learner journey, build the planned React web screens, implement accessible quiz interactions, and keep the visual hierarchy clear and consistent.
- **Justin-Yonardo** - mobile, backend, and quality owner: build the planned Expo applications, Hono API, and D1 schema; configure authentication and persistence; and coordinate automated and cross-platform testing.

Although each person will have a primary area, we plan to use shared code review, short integration checkpoints, and joint user testing. This will prevent one platform from drifting away from the common quiz format or learning goals.

Challenges We Expect to Run Into

Our most important challenge will be turning imperfect captions into questions that are accurate, answerable, and genuinely useful. Captions may be missing punctuation, contain transcription errors, or omit visual information shown in the lesson. AI output may also repeat concepts, invent details, or produce an explanation that conflicts with the stored answer. We plan to address these risks with source-grounded prompts, strict schemas, validation passes, duplicate checks, and targeted regeneration of only the invalid portion.

A second challenge will be coordinating asynchronous generation with a responsive interface. Learners should not wait for every question before they can begin, but the product must never score an incomplete quiz as though it were complete. We will need clear state transitions for loading, partial readiness, retries, completion, and failure. Cross-platform differences in browser extensions, native networking, secure storage, and authentication will add further complexity. Finally, we must balance privacy and convenience while ensuring that credentials never leave the user's device unintentionally.

Accomplishments We Are Proud Of

At this pre-development stage, we are proud of the product concept and the architectural decisions documented in our repository, not of software that has already been built. The planned innovation is the combination of a low-friction lesson-to-quiz workflow, early access to the first validated question, local control of the learner's AI credential, and a shared quiz contract across browser and mobile clients.

Compared with a basic quiz generator that returns a single block of unchecked text, ClipQuest is intended to behave like a complete learning product. It will ground questions in captions, validate structured output, preserve good questions when later ones fail, provide reasoned feedback, save progress, and support review over time. We are also proud that our plan treats error recovery, privacy, accessibility, and testing as core requirements rather than features to be added after a demo.

What We Learned

During planning, we learned that an AI response is only one part of a trustworthy educational experience. The system must manage source quality, question validity, grading consistency, waiting time, recovery, and storage as one connected workflow. We also learned that sharing schemas across clients can reduce mismatches, but only if ownership and versioning are clear. Privacy decisions shape architecture early: keeping the DeepSeek key on the device changes how generation, debugging, and support must work. Most importantly, we learned to separate an appealing idea from an implemented result. Honest status communication is essential, which is why this document explicitly states that no code has been written yet.

What Is Next for ClipQuest

Our next step will be to turn the current README into an executable foundation: define the monorepo, shared schemas, basic authentication, and one end-to-end Chrome path that can transform a captioned lesson into a completed quiz. We will then add robust validation and recovery, followed by saved lessons, progress tracking, and the native applications. Each milestone will require both automated checks and learner-visible testing.

If successful, ClipQuest could help students study more actively without requiring teachers to prepare a separate quiz for every public video. It could support independent learners, revision groups, and classrooms that already rely on online lessons. Given more time, we would add teacher-curated quiz sharing, multilingual caption and question support, spaced-repetition review, accessibility personalization, analytics that highlight weak concepts, and optional school integrations. We would also explore on-device or lower-cost models, stronger citation links from each question to the relevant caption moment, and evaluation datasets for measuring educational accuracy rather than judging generation by fluency alone.

PLANNED FIRST MILESTONE

One public, captioned YouTube lesson will produce a complete, validated quiz in Chrome; the learner will answer every question, receive coherent feedback, and see the finished attempt saved to their account.